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Anti–Glomerular Basement Membrane Disease and Goodpasture Disease

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The syndrome of kidney failure and lung hemorrhage was associated with the name of Ernest Goodpasture by Stanton and Tange in their description of nine cases in 1958.[1,2] All nine patients presented with lung hemorrhage and acute kidney failure and died within hours or days. These features had been prominent in the case of a young man who died during the influenza pandemic of 1919, whose postmortem findings were memorably reported by Goodpasture[1]: "The lungs gave the impression of having been injected with blood through the bronchi so that all the air spaces were filled" (Fig. 25.1).

Several diseases are now recognized as being associated with alveolar hemorrhage and rapidly progressive glomerulonephritis (RPGN). Nevertheless, this remains a striking clinical entity with relatively few causes and few pathogenetic mechanisms.

Because the first recognized mechanism was anti–glomerular basement membrane (anti-GBM) antibody formation and deposition, Goodpasture's name is firmly associated with anti-GBM disease (Goodpasture disease), even though this is responsible for only a proportion of patients with Goodpasture syndrome of lung hemorrhage and RPGN. The terminology used in this chapter is defined in Table 25.1.

ETIOLOGY AND PATHOGENESIS

Autoimmunity to a Component of Glomerular Basement Membrane

Goodpasture disease is caused by autoimmunity to the carboxyl terminal, noncollagenous (NC1) domain of a type IV collagen chain, α3(IV)NC1, also known as the Goodpasture antigen[3,4] (Fig. 25.2). Type IV collagen is an essential constituent of all basement membranes. In most tissues, it is composed of trimers including two α1 chains and one α2 chain, but there are also four tissue-specific chains, α3 through α6.[5,6] Three of these, α3 through α5, are found in GBM and in the basement membranes of the alveolus, the cochlea, parts of the eye (including corneal basement membrane and Bruch's membrane), the choroid plexus of the brain, and some endocrine organs.

All patients with RPGN, lung hemorrhage, and anti-GBM antibodies have antibodies to α3(IV)NC1, usually binding predominantly to a single or a very restricted set of epitopes. Some patients also have antibodies to other basement membrane constituents, including other collagen IV chains, usually in low titer.

Predisposing Factors

Both environmental and genetic factors appear to be important in etiology. There are strong associations between Goodpasture disease and human leukocyte antigen (HLA) class II alleles, including *DRB1∗1501* and *DR4* alleles, whereas *DR1* and *DR7* confer strong and dominant protection.[7] Some diseases and treatments predispose, as described in the following section.

Precipitating Factors

Theories of pathogenesis include precipitating factors that alter antigen processing to generate peptides that are usually destroyed or hidden, and to which tolerance is therefore deficient,[8,9] and molecular mimicry.[10] None of these is proved. Reports of temporal and geographic clustering of cases suggest an environmental trigger but no specific infectious agent has been consistently identified, leading to speculation that autoimmunity could be triggered (or amplified) by infection in general (or the response to infection), rather than the consequence of an aberrant immune response to a particular infectious agent.[11,12] Hydrocarbon exposure has been linked to disease onset in several striking case reports, but such exposure may simply trigger lung hemorrhage in patients who already have the disease. Furthermore, exposures of this kind are very common in the modern world. Similarly, cigarette smoking may precipitate lung hemorrhage in patients who already have circulating autoantibodies, but there is no evidence for a role in causation.

In several cases, kidney trauma or inflammation has preceded the development of the disease (Box 25.1). This may alter α3(IV)NC1 turnover and metabolism qualitatively or quantitatively, providing an opportunity for self-tolerance to be broken. Qualitative changes in the basement membrane epitopes presented to T cells could be a result of overloading of the usual or recruitment of alternative processing pathways, such as extracellular processing by proteases released into inflamed glomeruli. The quantity of α3(IV)NC1 presented to T cells may be greater where there has been damage to the basement membrane, as occurs in small-vessel vasculitis (see Chapter 26). Some features suggest that an anti-GBM response may be a secondary phenomenon in some patients with vasculitis.[13,14] The association with membranous nephropathy (MN) is interesting because the thickened GBM in that disease contains increased amounts of the tissue-specific type IV collagen chains, including the Goodpasture antigen. The same could apply to a possible association with long-standing type 1 diabetes mellitus.[15]

Mechanisms of Kidney Injury

The α3(IV)NC1 autoantibodies are central in the pathogenesis of Goodpasture disease[16,17] (Fig. 25.3). Antibodies eluted from the kidneys of patients who had died of Goodpasture disease rapidly bind to the GBM and cause glomerulonephritis (GN) when they are injected into monkeys.[18] The deposited antibodies are predominantly immunoglobulin G1 (IgG1). Contributions to kidney injury mediated by such antibodies come from complement and from neutrophil and macrophage infiltration. T cells are essential for driving autoantibody production by T cell–dependent B cells, and, in experimental kidney disease, they are critical in producing glomerular crescents,[16,19] which are a usual feature of Goodpasture disease. Moreover, in mice engineered to express the human susceptibility HLA allele *DRB1∗1501*, α3(IV)NC1-specific CD4 T cells are sufficient to transfer disease between animals.[20]

Agents that downregulate inflammation by inhibiting interleukin-1 (IL-1) or tumor necrosis factor (TNF), or that inhibit recruitment of inflammatory cells by blockade of adhesion molecules or chemoattractants, suppress injury in experimental models of anti-GBM disease. Evidence in humans and in experimental animals supports the severity of kidney injury being increased by proinflammatory cytokines or by stimuli likely to elicit them, such as bacteremia.[19] Crescent formation is seen in aggressive inflammatory GN, as described in Chapter 17 (see Fig. 17.7).

Lung Hemorrhage

Lung hemorrhage in Goodpasture disease (but not in small-vessel vasculitis, the other major cause of Goodpasture syndrome) occurs only if there is an additional insult to the lung, which is usually cigarette smoke. However, infection, fluid overload, toxicity from inhaled vapors or other irritants, and the systemic effects of some cytokines are also possibilities. The higher risk of kidney disease compared with lung hemorrhage may be because alveolar capillary endothelial cells more effectively block circulating immunoglobulin from reaching the underlying basement membrane. In the glomerulus, antibodies have more direct access to the GBM via the diaphragm-free fenestrations of glomerular endothelium. Other sites at which the Goodpasture antigen is found are not involved in Goodpasture disease, except possibly the choroid plexus, where the endothelium is again fenestrated, and more rarely the eye.

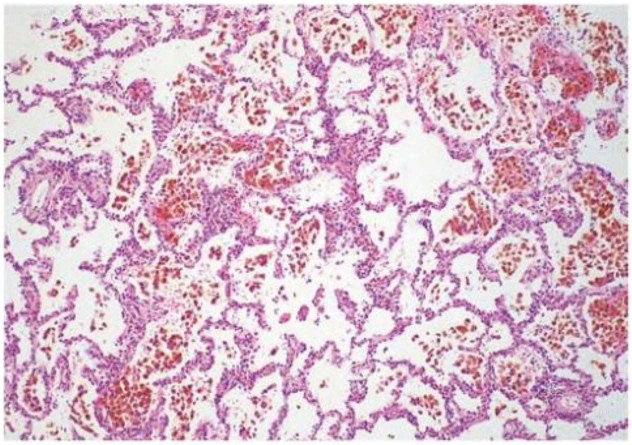

Fig. 25.1 Alveolar Hemorrhage. Open lung biopsy sample in a patient with Goodpasture disease shows alveolar hemorrhage. (Courtesy Dr. E. Mary Thompson, St Mary's Hospital, London.)

EPIDEMIOLOGY

Goodpasture disease is rare, with an incidence in both White and Chinese populations approaching 1 case per 1 million population per year.[15] The incidence in Black populations appears to be lower. The incidence in other racial groups is uncertain. There is a slight male predominance. Lung hemorrhage is more common in younger patients. Age incidence is bimodal, with peaks in third and sixth decades.[15]

CLINICAL MANIFESTATIONS

Between 50% and 75% of patients present with acute symptoms of lung hemorrhage and advanced kidney failure. Symptoms are usually confined to the preceding few weeks or months, but rapid progression (during days) or much slower progression (during many months) may occur. A lack of systemic symptoms, other than those related to anemia, is typical, although an apparently minor infection often triggers the clinical presentation.

Lung Hemorrhage

Lung hemorrhage may occur with kidney disease or in isolation. Presenting symptoms may include cough and hemoptysis, but lung hemorrhage may result in marked iron-deficiency anemia and exertional dyspnea, even in the absence of hemoptysis. Examination findings may include pallor, dry inspiratory crackles, signs of consolidation, or respiratory distress. Recent lung hemorrhage typically is shown on the radiograph as central shadowing that may traverse fissures and give rise to the appearance of an air bronchogram (Fig. 25.4). However, even lung hemorrhage sufficient to reduce the hemoglobin concentration may cause only minor or transient radiographic changes, and these cannot be distinguished radiologically from other causes of alveolar shadowing, notably edema or infection. The most sensitive indicator of recent lung hemorrhage is an increased uptake of inhaled carbon monoxide (DLCO). Patients with lung hemorrhage are usually current cigarette smokers.

In apparently isolated lung disease, progressive alveolar or fibrotic disease or pulmonary hemosiderosis may be suspected, although hematuria is usually found to be present if sought. This may continue for months or, in rare cases, recurrently for years before significant kidney disease occurs.

Glomerulonephritis

Patients with GN may notice dark or red urine, but progression to oliguria is sometimes so rapid that this phase, if it occurs, is missed. In a third to half of patients, GN occurs in the absence of lung hemorrhage.

TABLE 25.1 Definition of Terms Associated With Anti-GBM Disease and Goodpasture Syndrome

Term	Definition	Pathogenesis
Pulmonary-renal syndrome	Kidney and respiratory failure	Many causes (see Box 25.3)
Goodpasture syndrome	RPGN and alveolar hemorrhage	Several causes (see Box 25.4)
Anti-GBM disease	Disease associated with antibodies specific for (any) components of GBM	Most important are Goodpasture disease and Alport syndrome posttransplant anti-GBM disease
Goodpasture disease	Disease associated with autoantibodies specific for α3(IV)NC1 May include RPGN, lung hemorrhage, or both	Autoimmunity to α3(IV)NC1
Alport syndrome posttransplant anti-GBM disease	Glomerulonephritis associated with anti-GBM antibodies developing after kidney transplantation in patients with Alport syndrome	Immunity to foreign collagen IV chains not expressed in patients with Alport syndrome, usually α3 or α5(IV)NC1

GBM, Glomerular basement membrane; *RPGN,* rapidly progressive glomerulonephritis.

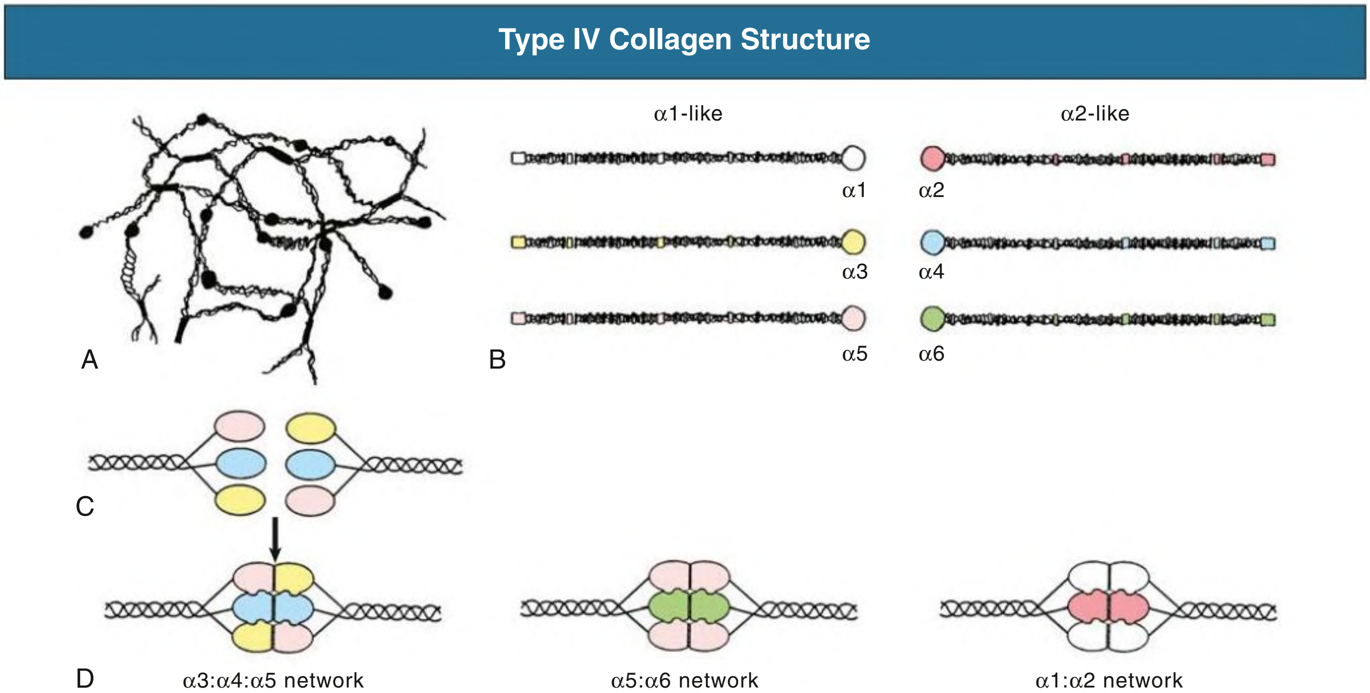

Fig. 25.2 Type IV Collagen Structure. (A) The type IV collagen network makes a "chicken wire" structure in the glomerular basement membrane (GBM). (B) Six paired type IV collagen genes, *COL4A1* to *COL4A6,* encode type IV collagen monomers α1 to α6. (C) These associate in two or three defined monomer types per protomer (carboxyl terminal domains of α3:α4:α5) to form three recognized networks (D). α1:α2 is present in almost all basement membranes; α3:α4:α5 is the major constituent of GBM and is a significant component of alveolar basement membrane and other locations; and α5: α6 is found in Bowman's capsule, skin, esophagus, and other locations.

BOX 25.1 Predisposing Events Associated With the Presentation of Goodpasture Disease

Possibly Induce Autoimmune Response and Disease
- Systemic small-vessel vasculitis affecting glomeruli
- Membranous nephropathy
- Lithotripsy of kidney stones
- Urinary obstruction
- Alemtuzumab therapy for multiple sclerosis

Precipitate Pulmonary Hemorrhage
- Cigarette smoke
- Hydrocarbon exposure
- Pulmonary infection
- Fluid overload

In this subgroup, because systemic symptoms are generally not prominent, presentation is often late with kidney failure.

Whatever the early pattern of disease, once significant kidney impairment has occurred, further deterioration in kidney function is usually rapid. Presentation at or shortly after acceleration of the disease process is common, and patients may demonstrate very rapid loss of kidney function and life-threatening lung hemorrhage. Urinalysis always reveals hematuria (even in apparently isolated pulmonary disease), usually modest proteinuria, and dysmorphic red blood cells (RBCs) and RBC casts on microscopy. The kidneys are generally of normal size but may be enlarged. Hematuria may be substantial or associated with loin pain in acute disease.

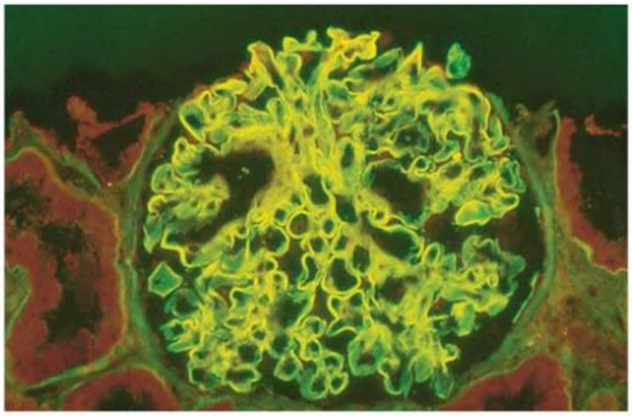

Fig. 25.3 Autoantibodies to Goodpasture Antigen Bound to a Normal Glomerulus. Direct immunofluorescence of normal kidney with sera from a patient with Goodpasture disease shows the antigen in a patient with lung hemorrhage and hematuria. (Courtesy Dr. Richard Herriot, Aberdeen Royal Infirmary, UK.)

PATHOLOGY

Kidney biopsy is essential because it provides diagnostic and prognostic information. Typical appearances are of diffuse proliferative GN with variable degrees of necrosis, crescent formation, glomerulosclerosis, and tubular loss (Fig. 25.5). The degree of crescent formation and tubular loss correlates with kidney prognosis. Characteristically, the crescents all appear to be of similar age and cellularity. When biopsy is performed earlier in the disease, changes may be limited to focal and segmental mesangial expansion, with or without necrosis.

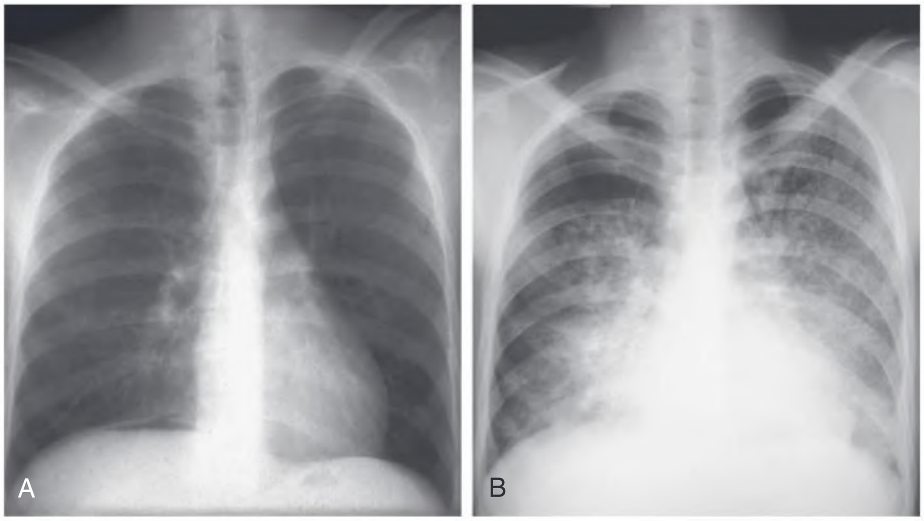

Fig. 25.4 Lung Hemorrhage. (A) Patient with early pulmonary hemorrhage. The chest radiograph still appears normal. (B) Radiograph taken 4 days later shows the evolution of alveolar shadowing caused by lung hemorrhage.

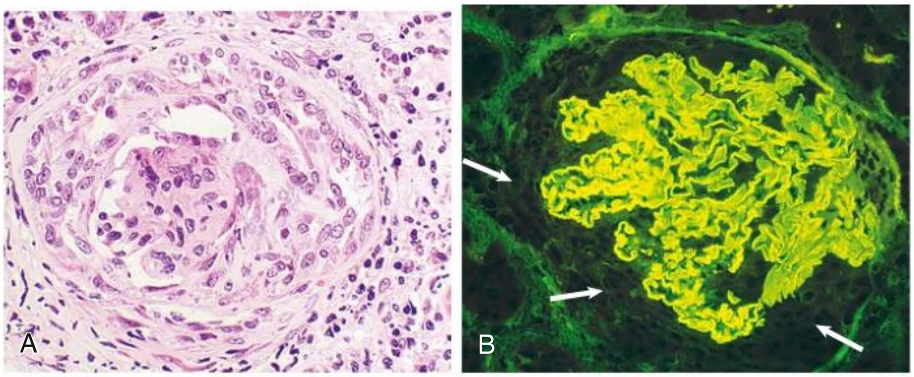

Fig. 25.5 Kidney Biopsy in Goodpasture Disease. (A) Glomerulus from a patient with Goodpasture disease showing a recent, mostly cellular crescent. (B) Direct immunofluorescence study showing ribbon-like linear deposition of immunoglobulin G along the glomerular basement membrane. The glomerular tuft is slightly compressed by cellular proliferation (exhibiting no immunofluorescence), forming a crescent *(arrows)*. (Courtesy Dr. Richard Herriot, Aberdeen Royal Infirmary, UK.)

This progresses to hypercellularity and then to more general changes, including fractures of the GBM and Bowman's capsule, neutrophils in the glomeruli, and glomerular capillary thrombosis.[21]

Immunohistology

In the presence of severe glomerular inflammation, linear deposition of Ig along the GBM is pathognomonic. The immunoglobulin is usually IgG, sometimes with IgA or IgM (10%–15%), but IgA alone is rarely detected. Linear deposition of C3 is detectable in about 75% of biopsies. Linear immunofluorescence (IF) with anti-immunoglobulin reagents is occasionally seen in other conditions, usually without glomerular inflammation (Box 25.2). In most such cases, the deposited immunoglobulin is less abundant than in Goodpasture disease and is either nonspecifically deposited or bound to GBM components other than type IV collagen chains.

Circulating IgG anti-GBM antibodies are almost invariably present and may be detected and quantified by use of immobilized Goodpasture antigen in an immunoassay. The titer of anti-GBM antibody at presentation correlates with the severity of nephritis, but sometimes is at a very low level despite significant disease; interpretation may require knowledge of the local laboratory's approach to classifying assay results. Treatment and relapse are often mirrored by changes in titer.

BOX 25.2 Conditions Associated With Linear Binding of Immunoglobulin to the GBM

Specific Binding to GBM

- Goodpasture syndrome
- Alport syndrome after kidney transplantation

Nonspecific Binding to GBM

- Diabetes
- Cadaver kidneys
- Light chain disease
- Fibrillary glomerulopathy
- Systemic lupus erythematosus (possibly specific but not considered directly pathogenic)

GBM, Glomerular basement membrane.

Pathology in Other Tissues

Pathologic changes in lung tissue can be difficult to interpret because the changes, including Ig deposition, are often patchy and may be missed.

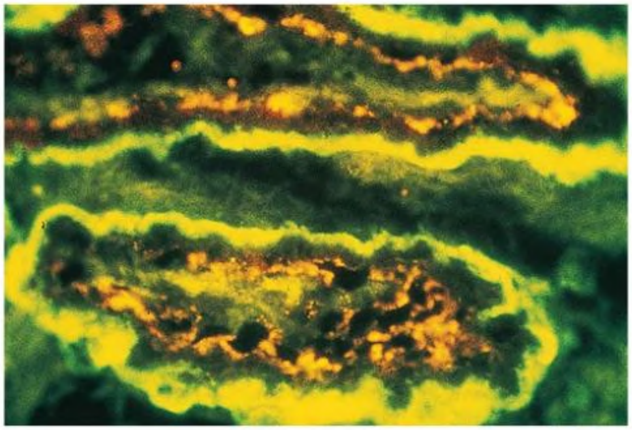

Fig. 25.6 Immunoglobulin G (IgG) Binding to Choroid Plexus. Direct immunofluorescence study showing binding of IgG to the choroid plexus of a patient who died of Goodpasture disease. (Courtesy Dr. Stephen Cashman, Imperial College, London.)

Frequently, the only findings are mild, chronic inflammation and hemosiderin-laden macrophages, which are consistent with other more common pathologic diagnoses. This makes negative bronchoscopic or open-lung biopsy findings unhelpful in excluding the diagnosis.

Other tissues in which α3(IV)NC1 is expressed are rarely available for pathologic analysis, but even if antibody is deposited in these other sites, it is rarely associated with clinical disease. A number of case reports describe neurologic syndromes, particularly seizures, that may be related to antibody deposition in the choroid plexus but may have other explanations in patients with acute kidney injury (Fig. 25.6). Other reports have described retinal detachment, in one case with antibody deposition, but again, this is rare. Placental tissue also contains the Goodpasture antigen and has been reported in a single case to act as a "sink" that binds anti-GBM antibody during pregnancy, resulting in exacerbation of GN after delivery.

DIFFERENTIAL DIAGNOSIS

Diagnosis of Goodpasture disease in patients who present with Goodpasture syndrome does not usually present difficulties once the possibility has been raised, although the urgency is often not appreciated. Direct IF on kidney tissue and assay for circulating anti-GBM antibodies are the most rapid techniques, and kidney biopsy is always indicated. Diagnosis is often delayed when patients present with subacute disease affecting the lung or the kidney in isolation. Patients with subacute lung hemorrhage may not report hemoptysis and may present with diffuse lung disease, which has many causes. Dipstick testing for hematuria is important.

Detection of Anti–Glomerular Basement Membrane Antibodies

Direct immunohistology is very sensitive for detection of anti-GBM antibody production because the GBM selectively adsorbs and concentrates low levels of circulating antibody. However, in some circumstances, GBM may also adsorb antibody nonspecifically (see Box 25.2). Detection of anti-GBM antibodies in serum requires immunoassays based on preparations of human or animal GBM or recombinant antigen. The performance of these assays is variable. False negatives may be encountered with low titers of antibody (typically in apparently isolated pulmonary disease) and, rarely, with antibodies targeting unusual epitopes or unusual types (e.g., IgA or monoclonal proteins). Confirmation of the specificity of anti-GBM antibodies may be obtained by Western blotting of serum onto solubilized human GBM or recombinant α3(IV)NC1, usually at a reference laboratory. Indirect immunohistology (putting patient serum onto normal kidney sections) is too insensitive for reliable diagnostic use.

False-positive results may be encountered in sera from patients with inflammatory diseases that often exhibit increased nonspecific binding. This places greater emphasis on the purity of antigen used for anti-GBM assays. False-negative results are usually encountered in patients with low titers of antibodies in association with isolated lung disease or with very early or subacute kidney disease. Low titers also may be associated with anti-GBM disease that occurs after kidney transplantation in patients with Alport syndrome (see later discussion).

In very advanced disease, linear antibody deposition may not be seen because of extensive destruction of GBM structure. Otherwise, deposited Ig remains detectable for some months after immunoassays have become negative.

Patients With Anti-GBM Antibodies and Other Diseases

Antineutrophil Cytoplasmic Antibody and Systemic Small-Vessel Vasculitis

Anti-GBM antibodies are sometimes detected in patients with antineutrophil cytoplasmic antibodies (ANCAs), especially ANCAs with specificity for myeloperoxidase (see Chapter 26). Such "double-positive" patients may have a clinical course and response to treatment more typical of vasculitis than of Goodpasture disease and have possibly developed anti-GBM antibodies secondary to vasculitic glomerular damage.[8–11] Anti-GBM titers tend to be lower in ANCA-positive anti-GBM antibody–positive patients than in patients with anti-GBM antibodies alone. Recovery of kidney function may be more likely if ANCAs are present, even if patients are dialysis dependent when treatment is started, although newer series have failed to detect the differences described in early reports.

Membranous Nephropathy

Anti-GBM antibodies are occasionally identified in patients with MN, usually coincident with an accelerated decline in kidney function and the formation of glomerular crescents.[5,15,22] About two-thirds of studies report evidence of evolution from preexisting nephrotic syndrome, and about half report that a previous kidney biopsy showed typical MN. Progression to end-stage kidney disease (ESKD) has usually been rapid, but the diagnosis has rarely been made at a stage early enough to expect intensive treatment to be successful. Three patients with Goodpasture disease later developed typical MN.

Alemtuzumab Treatment

Treatment of multiple sclerosis (MS) with alemtuzumab, a monoclonal antibody targeting CD52 on B and T cells, is associated with the development of new autoimmunity in approximately 30% of patients, including rare cases of anti-GBM disease, sometimes as late as 4 years after treatment.[23] It remains to be established whether carriage of *DRB1*1501,* which is overrepresented in MS, influences the risk for developing anti-GBM disease after alemtuzumab therapy.

Pulmonary-Renal Syndromes

A wide variety of conditions may cause simultaneous pulmonary and kidney disease. The term *pulmonary-renal syndrome* implies failure of both organs, the most common cause being fluid overload in a patient with kidney failure of any cause (Box 25.3). Then there are diseases associated with pulmonary hemorrhage and RPGN, sometimes called Goodpasture syndrome (Box 25.4).

BOX 25.3 Nonimmune Causes of Pulmonary-Renal Syndrome

With Pulmonary Edema
- Acute kidney injury with hypervolemia
- Severe cardiac failure

Infective
- Severe bacterial pneumonia (e.g., *Legionella*) with kidney failure
- Hantavirus infection
- Opportunistic infections in the immunocompromised patient

Other
- Acute respiratory distress syndrome with kidney failure in multiorgan failure
- Paraquat poisoning
- Renal vein/inferior vena cava thrombosis with pulmonary emboli

BOX 25.4 Causes of Lung Hemorrhage and Rapidly Progressive Glomerulonephritis

Diseases Associated With Antibodies to the GBM (20%–40% of Cases)
- Goodpasture disease (spontaneous anti-GBM disease)

Diseases Associated With Systemic Vasculitis (60%–80% of Cases)
- Granulomatosis with polyangiitis (common)
- Microscopic polyangiitis
- Systemic lupus erythematosus
- Eosinophilic granulomatosis with polyangiitis (Churg-Strauss)
- IgA vasculitis (Henoch-Schönlein purpura)
- Henoch-Schönlein purpura
- Behçet syndrome
- Essential mixed cryoglobulinemia
- Rheumatoid vasculitis
- Drugs: Penicillamine, hydralazine, propylthiouracil

GBM, Glomerular basement membrane; *IgA*, immunoglobulin A.

The two classes of disease in Box 25.4 can sometimes be differentiated clinically, but serology and kidney biopsy are usually required. Kidney biopsy also provides valuable prognostic information.

NATURAL HISTORY

There is some variability in the pattern of early disease. Most patients present acutely with lung hemorrhage or advanced kidney failure and report that the illness developed over weeks or a few months. However, there are several reports of patients presenting with mild respiratory symptoms or incidental microhematuria, with disease progressing much more slowly during months or years; some have abruptly developed the full acute syndrome. Microhematuria preceded kidney failure in all the patients who developed anti-GBM disease while being monitored after alemtuzumab treatment of MS. A study of US veterans showed a similarly long delay between first appearance of antibodies and clinical presentation.[24]

Once RPGN has developed, kidney function is rapidly and often irretrievably lost. Progression is often much more rapid than in RPGN occurring in other contexts, such as microscopic polyangiitis, perhaps because more glomeruli are simultaneously affected. Consequently, there is a much narrower window of opportunity for effective treatment than when other diseases cause the syndrome.

Although a severe exacerbation of lung disease usually coincides with deterioration of kidney function, the natural history of isolated lung disease critically depends on continued exposure to pulmonary irritants.

TREATMENT

Immunosuppressive Regimens

Before the introduction of immunosuppressive treatment, most patients died shortly after the development of renal impairment or lung hemorrhage.[19] Lung hemorrhage now usually can be arrested within days. Kidney function can be protected if impairment is mild, and even severe kidney impairment can be reversed in some patients. However, dialysis-dependent patients rarely recover kidney function despite immunosuppression and should probably be immunosuppressed only if lung hemorrhage occurs.

Fig. 25.7 shows a chart recording the treatment of a patient with Goodpasture disease. Treatment recommendations for acute severe disease were devised to reduce levels of circulating pathogenic antibodies as rapidly as possible and to lessen their contribution to rapid glomerular destruction (Table 25.2). However, this regimen is almost certainly effective through a much broader range of mechanisms, including T-cell depletion. Once the disease is controlled, immunosuppression usually can be tapered off over 3 months, and subsequent relapse is uncommon. The immune response is self-limited in the absence of immunosuppression, with antibodies disappearing over 1 to 2 years. Spontaneous remissions and effectiveness of relatively brief periods of immunosuppression are in striking contrast to the more prolonged immunosuppression generally required to prevent relapse of vasculitis and suggest a greater capacity for restoration of usual tolerance to α3(IV)NC1[25] than to targets in vasculitis (see Chapter 26).

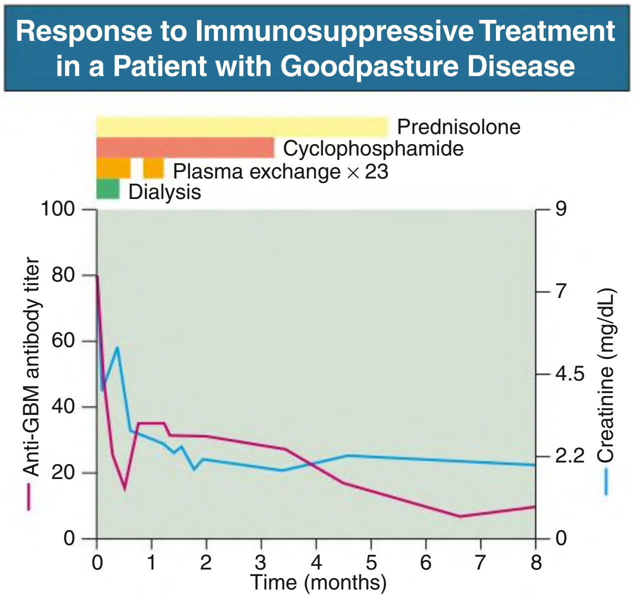

Fig. 25.7 **Response to Immunosuppressive Treatment in Goodpasture Disease.** The patient required dialysis but had no lung hemorrhage. The good response to treatment was unusual but not unique. The kidney biopsy showed that 85% of glomeruli contained recent (mostly cellular) crescents, suggesting very acute disease, which may be indicative of a more favorable response to treatment. *GBM*, Glomerular basement membrane.

TABLE 25.2 **Treatment Regimen for Acute Goodpasture Disease**

Therapy	Recommendation
Prednisolone	1 mg/kg/24 hr orally. Reduce at weekly intervals to achieve one-sixth of this dose by 8 wk. For a starting daily dose of 60 mg, use weekly reductions to 45, 30, 25, 20, and 15 mg; then alternate weekly reductions to 12.5 and 10 mg. Maintain this dose to 3 months; then taper to stop by 4 months.
Cyclophosphamide	3 mg/kg/24 h orally, rounded down to the nearest 50 mg. Patients >55 years receive a reduced dose of 2.5 mg/kg. Discontinue after 3 months.
Plasma exchange	Daily exchange of 1 volume of plasma for 5% human albumin for 14 days or until the circulating antibody is suppressed. In the presence of pulmonary hemorrhage or within 48 hours of invasive procedure, 300–400 mL of fresh-frozen plasma is given at end of each treatment or according to coagulation tests.
Monitoring	Daily blood count during plasma exchange and while antibody titer remains elevated. At least twice weekly during first month, weekly thereafter. If white blood cell count decreases to $<3.5 \times 10^9$/L, stop cyclophosphamide until the count recovers. Resume at lower dose if cessation has been necessary. Baseline D_{LCO}, with further measurements as indicated. Daily coagulation tests during plasma exchange to monitor for significant depletion of clotting factors. Initially, daily checks of kidney and hepatic function and glucose.
Prophylaxis against complications of treatment	Oral antifungal lozenges or rinse; proton pump inhibitor. Cotrimoxazole prophylaxis against *Pneumocystis jirovecii*. Avoid nonessential lines and catheters.

D_{LCO}, Diffusing capacity of lung for carbon monoxide.

In RPGN with no evidence of an infective cause, immunosuppressive therapy should be started immediately, sometimes before the kidney biopsy findings are available. If therapy is stopped after a few days, the patient will have incurred minimal risk (as long as pulse high-dose corticosteroids are avoided) but may have much to gain from earlier treatment.

Plasma Exchange and Immunosuppression

The regimen described in Table 25.2 dramatically improved the outlook for patients when it was introduced in the 1970s. An early randomized trial suggested some additional benefit of plasma exchange, but the interpretation was complicated by the recipient group's less severe disease at presentation.[26] It showed that milder disease can be effectively treated with corticosteroids and cyclophosphamide alone, although the overall outcomes for all patients were not as good as described with more intensive regimens.[26] Historical evidence suggests that treatment with corticosteroids alone, or corticosteroids with azathioprine, is less effective. Plasma exchange is of value only if it is accompanied by adjunctive immunosuppressive therapy.

Plasma exchange is likely to benefit kidney outcome by removing circulating auto-antibody so there is interest in new treatments that are more specific or more potent in their capacity to remove autoantibody. More specific is immunoadsorption to protein A, which also lowers anti-GBM antibodies rapidly and does not deplete complement components or clotting factors: A few reports suggest that it is as effective as plasma exchange. More effective is treatment with IdeS, a bacterial IgGase, which has been shown to rapidly and profoundly deplete circulating IgG anti-GBM autoantibody in patients with refractory anti-GBM disease.[27] The place of IdeS in therapy is yet to be worked out, but on current understanding of the immunopathogenesis of anti-GBM disease, it could well replace plasma exchange offering more rapid, more complete, and more selective depletion of circulating anti-GBM IgG and possibly reducing the deleterious effects of autoantibody that is already bound to the GBM.[28]

Information is lacking on the effectiveness of newer immunosuppressive agents such as mycophenolate mofetil (MMF) or anti–B cell antibodies, which tend to have only a slow effect on antibody production but may affect antigen presentation. Therefore, it is difficult to justify their use over proven therapy in the acute phase of this often rapidly progressive disease, but there may be a role in particular circumstances. Rituximab has anecdotally led to reductions in antibody titer in patients with autoimmune responses persisting or relapsing after usual therapy.[29]

Lung hemorrhage occurring alone tends to be relapsing and remitting, so there have been many reports of treatments (e.g., bilateral nephrectomy) that may help. Pulse methylprednisolone has been advocated, but high doses of corticosteroids fail to alter the underlying pathogenetic immune response and put the patient at increased risk for infectious and other complications. We recommend treating seriously ill patients with moderate doses of corticosteroids plus plasma exchange and cyclophosphamide. In contrast to advanced kidney failure, in which treatment is unlikely to lead to recovery of kidney function, even severe lung hemorrhage is likely to respond to treatment with full or almost full recovery of lung function.

In other acute severe diseases, daily administration of cyclophosphamide often has been superseded by pulse administration. We still prefer to use daily oral administration because it is known to work and requires only 3 months of therapy. Patients unable to take the drug orally can be given daily intravenous therapy at the usual oral dose. Dose does not need to be reduced in severe kidney failure, provided the white blood cell (WBC) count is monitored closely, but reductions for older patients are important (see Table 25.2) and close monitoring of leukocyte counts is imperative in all patients. If pulsed therapy were chosen, the CYCLOPS (Randomised Trial of Daily Oral Versus Pulse Cyclophosphamide as Therapy for ANCA-Associated Systemic Vasculitis) regimen would be a reasonable, if untested, option (see Chapter 26).

Results from all series show that recovery of kidney function is unlikely if, at initiation of treatment, the patient is oliguric, has a very high proportion of glomeruli with circumferential crescents, or has a serum creatinine level greater than 5.5 to 6.5 mg/dL (~500–600 μmol/L).[30] This is a notably different experience from that encountered in systemic vasculitis or idiopathic RPGN (see Chapter 26), in which kidney disease of apparently similar severity (using histology and serum creatinine) can be salvaged by similar treatment protocols.[31] This has led to the suggestion that immunosuppressive treatment should be withheld from patients with only slight chance of recovery (Table 25.3).

Supportive Treatment

The most likely cause of death in the first few days is respiratory failure caused by lung hemorrhage. Lung hemorrhage may be precipitated or exacerbated by the following:

- Fluid overload
- Smoking and other pulmonary irritants, possibly including high inspired oxygen concentration (F_{IO_2})
- Local or distant infection

TABLE 25.3 Factors to Consider in Aggressive Treatment of Goodpasture Disease

	Factors Favoring	Factors Against
Pulmonary hemorrhage	Present	Absent
Oliguria	Absent	Present
Creatinine	<5.5 mg/dL (~500 µmol/L)	>5.5–6.5 mg/dL (~500–600 µmol/L) and ANCA negative Severe damage on kidney biopsy No desire for early kidney transplantation
Other factors	Creatinine >5.5–6.5 mg/dL (~500–600 µmol/L) *but* rapid and recent progression ANCA positive Glomerular damage less severe than expected Crescents recent, nonfibrous Early kidney transplantation desired	
Associated disease	Absent	Unusually high risk from immunosuppression

ANCA, Antineutrophil cytoplasmic antibody.

- Anticoagulation used during dialysis or plasma exchange
- Thrombocytopenia, defibrination, and depletion of clotting factors as a consequence of plasma exchange

It is therefore advisable to ensure correct fluid balance, to prohibit smoking, to use the lowest fractional FiO_2 that gives adequate oxygenation, and to minimize the use of heparin.

Plasma exchange (see Table 25.2) should be monitored by daily blood counts, calcium concentration (if regional citrate anticoagulation is used), and coagulation tests. Diminished clotting factor levels should be replenished by administration of fresh-frozen plasma or clotting factor preparations at the end of each plasma exchange session, as required.

After the first few days, the major cause of morbidity and mortality is infection. Infection carries the added risk for potentiating glomerular and lung inflammation and injury, so precautions to reduce risk, such as minimizing indwelling cannulas, are important. If leukopenia less than 3.5×10^9/L or neutropenia develops, cyclophosphamide should be discontinued and resumed at a lower dose when the neutrophil count recovers, if necessary with the assistance of granulocyte colony-stimulating factor.

Monitoring Effect of Treatment on Disease Activity

The effect of treatment on the kidney disease is monitored by following serum creatinine values. Indicators of recent lung hemorrhage include hemoptysis, decreases in hemoglobin concentration, chest radiograph changes, and increases in the DLCO, with the last being the most sensitive. Any worsening of symptoms during treatment may indicate inadequate immunosuppression, but it is frequently a consequence of intercurrent infection exaggerating immunologic injury or fluid overload or other factors precipitating lung hemorrhage.

Monitoring of anti-GBM titers during, and particularly 24 hours after, the last planned plasma exchange treatment is useful for confirming effective suppression of autoantibodies. They should be undetectable within 8 weeks, but even without treatment, autoantibodies generally become undetectable by an average of 14 months.

Duration of Treatment and Relapses

Corticosteroid treatment may be gradually reduced and cyclophosphamide discontinued at 3 months. In contrast to treatment of small-vessel vasculitis, immunosuppression longer than this is usually not necessary. Longer treatment is appropriate for patients who are positive for both anti-GBM antibody and ANCA (see later discussion). Late increases in anti-GBM level may predict clinical relapse, although antibodies are generally permanently suppressed in patients who have completed the immunosuppressive regimen. If there is recurrence, success has been achieved by treating as at first presentation.

Electing Not to Treat

Advanced kidney failure, frequently already established at presentation, is generally not salvaged by any current treatment.[30,32,33] Furthermore, the immunosuppressive regimen outlined carries significant risks and careful monitoring is required. For these reasons, it may be reasonable not to initiate immunosuppression in patients who present with advanced kidney failure without lung hemorrhage. The decision not to treat is strengthened if the kidney biopsy specimen shows widespread glomerulosclerosis and tubular loss and the patient is dialysis dependent at presentation (see Table 25.3). The risk for development of late lung hemorrhage in these patients seems to be low but warrants particular care to avoid the major precipitating factors, smoking and pulmonary edema, in at least the first few months. However, patients who are dialysis dependent usually should be treated if the kidney histopathologic changes are unexpectedly mild or very recent (highly cellular crescents, even if 100% of glomeruli are involved, or acute tubular necrosis). Several reports describe good outcomes in these patients even after prolonged oliguria.

Treatment of Double-Positive Patients

Patients with both ANCA and anti-GBM antibodies may have other extrarenal disease requiring treatment (see Table 25.3). There is conflicting evidence as to whether their kidney prognosis is the same as or better than that of other patients with anti-GBM antibodies. Earlier series suggested a better prognosis, but this was not confirmed in two later reports.[13,14] Because of the risk for serious disease in other organs, double-positive patients should usually receive an immunosuppressive regimen similar to that given for small-vessel vasculitis, with continuing immunosuppression with azathioprine after 3 months of cyclophosphamide (see Chapter 26).

TRANSPLANTATION

Kidney transplantation in patients who have had Goodpasture disease carries the additional risk for disease recurrence. Recurrence with consequent loss of the graft has been reported and appears more likely when circulating anti-GBM antibodies are still detectable at transplantation. Therefore, it is reasonable to delay transplantation until circulating anti-GBM antibodies have been undetectable for 6 months and to monitor graft function, urinary sediment, and circulating anti-GBM antibody levels to detect recurrent disease (see Chapter 113). Biopsy samples of well-functioning grafts sometimes show linear deposition of Ig on the GBM without clinical or histologic disease or apparently an adverse prognosis.

ALPORT SYNDROME POSTTRANSPLANT ANTI–GLOMERULAR BASEMENT MEMBRANE DISEASE

Patients with Alport syndrome have mutations in a gene encoding one of the tissue-specific type IV collagen chains, usually α5. Because these chains assemble with each other during biosynthesis, the resulting phenotype in the case of most mutations often has all the tissue-specific chains (α3–α5) missing from the basement membranes, where they are normally coexpressed. Altered expression may lead to absent or inadequate immunologic tolerance to these proteins and to preservation of the capacity to mount a powerful (allo)immune response to the type IV collagen chains expressed in a normal donor kidney after kidney transplantation. Most patients with Alport syndrome accommodate kidney transplants with conventional immunosuppression without development of anti-GBM nephritis. However, development of low titers of anti-GBM antibodies is shown by many such patients having linear deposition of IgG on the GBM of the transplanted kidney on direct IF, without disease. This alone does not justify treatment.

Up to 2% of patients with Alport syndrome have been reported to develop RPGN in the transplanted kidney, although anecdotally this is much less common than historically. It is clinically indistinguishable from spontaneous anti-GBM disease but without lung hemorrhage. This is more likely if the patient has a large gene deletion causing the disease rather than a point mutation, with the inference that the immune system has never been exposed to the mature protein. Typically, graft function is lost despite treatment for presumed acute rejection. Disease is usually encountered some months or longer after a first kidney transplant, after weeks in a second, and after days in a third.[32] However, regrafting has been successful in two cases known to us and in two further cases in the literature. If the disease is recognized early, there are sound theoretical reasons for treating with the regimen recommended for Goodpasture disease, but there are few data on its effectiveness.[34]

In contrast to spontaneous Goodpasture disease, the specificity of anti-GBM antibodies in Alport syndrome posttransplant anti-GBM disease is not always to α3(IV)NC1. In many patients, possibly in most, the autoantibodies are specific for α5(IV)NC1, encoded by the *COL4A5* gene usually implicated in causation of the disease.[35] This is important because immunoassays for anti-GBM antibodies are optimized for detection of the anti-α3(IV)NC1 antibodies of spontaneous Goodpasture disease, and they may have low sensitivity for anti-α5(IV)NC1 antibodies. In the absence of widely available assays for these uncommon antibodies, kidney biopsy with immunohistology is the only reliable method of diagnosis.

SELF-ASSESSMENT QUESTIONS

1. A 65-year-old man presenting with acute kidney failure and pulmonary hemorrhage was diagnosed after appropriate investigation as having Goodpasture disease and was treated with plasma exchange (10 × 4 L over 2 weeks), oral cyclophosphamide (150 mg/day), and prednisolone (week 1: 60 mg/day; week 2: 45 mg/day; week 3: 30 mg/day; week 4: 25 mg/day; week 5: 20 mg/day; week 6: 15 mg/day; weeks 7–8: 20 mg alternate days). Cyclophosphamide (CYP) had to be omitted for 5 days, then restarted at 100 mg/day in week 4 because of neutropenia. At review 3 months after treatment commenced, the patient is entirely well with estimated glomerular filtration rate of 36 mL/min, hemoglobin of 110 g/L, and white blood cell count of 2.4×10^3 per µL. Anti-GBM antibodies remain detectable just above the reporting threshold of the local assay. How should his immunosuppression be managed?
   - **A.** Replace CYP with azathioprine or MMF and maintain with steroids to at least 1 year.
   - **B.** Discontinue CYP and steroids with close outpatient monitoring.
   - **C.** Continue CYP at a reduced dose with prednisolone (10 mg/day) or equivalent to at least 1 year.
   - **D.** Reinstitute plasmapheresis.
2. A 45-year-old nonsmoking executive collapses at a meeting and in the emergency department is found to have a serum creatinine of 1800 µmol/L, hemoglobin of 90 g/L, potassium of 7.2 mmol/L, blood pressure (BP) 165/92 mm Hg, and normal-sized unobstructed kidneys on ultrasound scan. After appropriate acute dialysis and BP control, a kidney biopsy sample is taken, which shows severe crescentic GN affecting 32/32 sampled glomeruli. On silver staining, breaks could be seen in multiple capillary loops and in Bowman's capsule of most glomeruli. Linear deposition of IgG along the residual GBM and strong positivity for serum anti-GBM antibodies establishes a diagnosis of Goodpasture disease. ANCA testing is negative, and there is no evidence of lung hemorrhage. What treatment would you recommend for this patient with Goodpasture disease?
   - **A.** Plasma exchange, CYP, and oral prednisolone for at least 3 months
   - **B.** Plasma exchange, CYP, and oral prednisolone with plan for early discontinuation in the event of continuing dependence on dialysis
   - **C.** Close monitoring with a goal of avoiding immunosuppression
   - **D.** CYP and oral prednisolone for at least 3 months
3. A 26-year-old man has a deceased donor renal transplant after 10 months on hemodialysis. His kidney failure is of unknown cause. Eight months posttransplant, kidney function is found to have deteriorated dramatically, with creatinine rising from 130 to 800 µmol/L at routine clinic visits 3 weeks apart. A biopsy shows 100% crescentic nephritis with strong linear binding of IgG to the GBM. Serum anti-GBM titers are negative. The most likely diagnosis is:
   - **A.** Anti-GBM (Goodpasture) disease
   - **B.** Alport anti-GBM disease
   - **C.** Antibody-mediated rejection
   - **D.** Polyomavirus infection

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